

4-12 Intensities of powder pattern lines. We are now in a position to gather together the factors discussed in preceding sections into an equation for the relative intensity of powder pattern lines:

$$I = |F|^2 p \left(\frac{1 + \cos^2 2\theta}{\sin^2 \theta \cos \theta} \right), \quad (4-12)$$

where I = relative integrated intensity (arbitrary units), F = structure factor, p = multiplicity factor, and θ = Bragg angle. In arriving at this equation, we have omitted factors which are constant for all lines of the pattern. For example, all that is retained of the Thomson equation (Eq. 4-2) is the polarization factor $(1 + \cos^2 2\theta)$, with constant factors, such as the intensity of the incident beam and the charge and mass of the electron, omitted. The intensity of a diffraction line is also directly proportional to the irradiated volume of the specimen and inversely proportional to the camera radius, but these factors are again constant for all diffraction lines and may be neglected. Omission of the temperature and absorption factors means that Eq. (4-12) is valid only for the Debye-Scherrer method and then only for lines fairly close together on the pattern; this latter restriction is not as serious as it may sound. Equation (4-12) is also restricted to the Debye-Scherrer method because of the particular way in which the Lorentz factor was determined; other methods, such as those involving focusing cameras, will require a modification of the Lorentz factor given here. In addition, the individual crystals making up the powder specimen must have completely random orientations if Eq. (4-12) is to apply. Finally, it should be remembered that this equation gives the relative *integrated* intensity, i.e., the relative area under the curve of intensity *vs.* 2θ .

It should be noted that "integrated intensity" is not really *intensity*, since intensity is expressed in terms of energy crossing unit area per unit of time. A beam diffracted by a powder specimen carries a certain amount of energy per unit time and one could quite properly refer to the total *power* of the diffracted beam. If this beam is then incident on a measuring device, such as photographic film, for a certain length of time and if a curve of diffracted intensity *vs.* 2θ is constructed from the measurements, then the area under this curve gives the total *energy* in the diffracted beam. This is the quantity commonly referred to as integrated intensity. A more descriptive term would be "total diffracted energy," but the term "integrated intensity" has been too long entrenched in the vocabulary of x-ray diffraction to be changed now.

4-13 Examples of intensity calculations. The use of Eq. (4-12) will be illustrated by the calculation of the position and relative intensities of

the diffraction lines on a powder pattern of copper, made with $\text{Cu } K\alpha$ radiation. The calculations are most readily carried out in tabular form, as in Table 4-2.

TABLE 4-2

1	2	3	4	5	6	7	8
Line	hkl	$h^2 + k^2 + l^2$	$\sin^2 \theta$	$\sin \theta$	θ	$\frac{\sin \theta}{\lambda} (\text{\AA}^{-1})$	f_{Cu}
1	111	3	0.1365	0.369	21.7°	0.240	20.0
2	200	4	0.1820	0.426	25.2	0.277	18.7
3	220	8	0.364	0.602	37.0	0.391	15.6
4	311	11	0.500	0.706	44.9	0.459	14.0
5	222	12	0.546	0.738	47.6	0.479	13.7
6	400	16	0.728	0.851	58.3	0.553	12.4
7	331	19	0.865	0.929	68.3	0.604	11.7
8	420	20	0.910	0.951	72.0	0.618	11.4

1	9	10	11	12	13	14
Line	F^2	p	$\frac{1 + \cos^2 2\theta}{\sin^2 \theta \cos \theta}$	Relative integrated intensity		
				Calc.	Calc.	Obs.
1	6400	8	12.10	6.20 × 10 ⁵	10.0	s
2	5600	6	8.50	2.86	4.6	m
3	3890	12	3.75	1.75	2.8	m
4	3140	24	2.87	2.16	3.5	s
5	3000	8	2.75	0.66	1.1	w
6	2460	6	3.18	0.47	0.8	w
7	2190	24	4.75	2.50	4.0	vs
8	2080	24	5.92	2.96	4.8	vs

Remarks:

Column 2: Since copper is face-centered cubic, F is equal to $4f_{\text{Cu}}$ for lines of unmixed indices and zero for lines of mixed indices. The reflecting plane indices, all unmixed, are written down in this column in order of increasing values of $(h^2 + k^2 + l^2)$, from Appendix 6.

Column 4: For a cubic crystal, values of $\sin^2 \theta$ are given by Eq. (3-10):

$$\sin^2 \theta = \frac{\lambda^2}{4a^2} (h^2 + k^2 + l^2).$$

In this case, $\lambda = 1.542\text{\AA}$ ($\text{Cu } K\alpha$) and $a = 3.615\text{\AA}$ (lattice parameter of copper). Therefore, multiplication of the integers in column 3 by $\lambda^2/4a^2 = 0.0455$ gives the values of $\sin^2 \theta$ listed in column 4. In this and similar calculations, slide-rule accuracy is ample.

Column 6: Needed to determine the Lorentz-polarization factor and $(\sin \theta)/\lambda$.

Column 7: Obtained from Appendix 7. Needed to determine f_{Cu} .

Column 8: Read from the curve of Fig. 4-6.

Column 9: Obtained from the relation $F^2 = 16f_{\text{Cu}}^2$.

Column 10: Obtained from Appendix 9.

Column 11: Obtained from Appendix 10.

Column 12: These values are the product of the values in columns 9, 10, and 11.

Column 13: Values from column 12 recalculated to give the first line an arbitrary intensity of 10.

Column 14: These entries give the observed intensities, visually estimated according to the following simple scale, from the pattern shown in Fig. 3-13(a) (vs = very strong, s = strong, m = medium, w = weak).

The agreement obtained here between observed and calculated intensities is satisfactory. For example, lines 1 and 2 are observed to be of strong and medium intensity, their respective calculated intensities being 10 and 4.6. Similar agreement can be found by comparing the intensities of any pair of neighboring lines in the pattern. Note, however, that the comparison must be made between lines which are not too far apart: for example, the calculated intensity of line 2 is greater than that of line 4, whereas line 4 is observed to be stronger than line 2. Similarly, the strongest lines on the pattern are lines 7 and 8, while calculations show line 1 to be strongest. Errors of this kind arise from the omission of the absorption and temperature factors from the calculation.

A more complicated structure may now be considered, namely that of the zinc-blende form of ZnS, shown in Fig. 2-19(b). This form of ZnS is cubic and has a lattice parameter of 5.41Å. We will calculate the relative intensities of the first six lines on a pattern made with Cu $K\alpha$ radiation.

As always, the first step is to work out the structure factor. ZnS has four zinc and four sulfur atoms per unit cell, located in the following positions:

Zn: $\frac{1}{4} \frac{1}{4} \frac{1}{4} +$ face-centering translations,

S: 0 0 0 + face-centering translations.

Since the structure is face-centered, we know that the structure factor will be zero for planes of mixed indices. We also know, from example (e) of Sec. 4-6, that the terms in the structure-factor equation corresponding to the face-centering translations can be factored out and the equation for unmixed indices written down at once:

$$F = 4[f_s + f_{\text{Zn}}e^{(\pi i/2)(h+k+l)}].$$

$|F|^2$ is obtained by multiplication of the above by its complex conjugate:

$$|F|^2 = 16[f_s + f_{\text{Zn}}e^{(\pi i/2)(h+k+l)}][f_s + f_{\text{Zn}}e^{-(\pi i/2)(h+k+l)}].$$

This equation reduces to the following form:

$$|F|^2 = 16 \left[f_s^2 + f_{\text{Zn}}^2 + 2f_s f_{\text{Zn}} \cos \frac{\pi}{2} (h + k + l) \right].$$